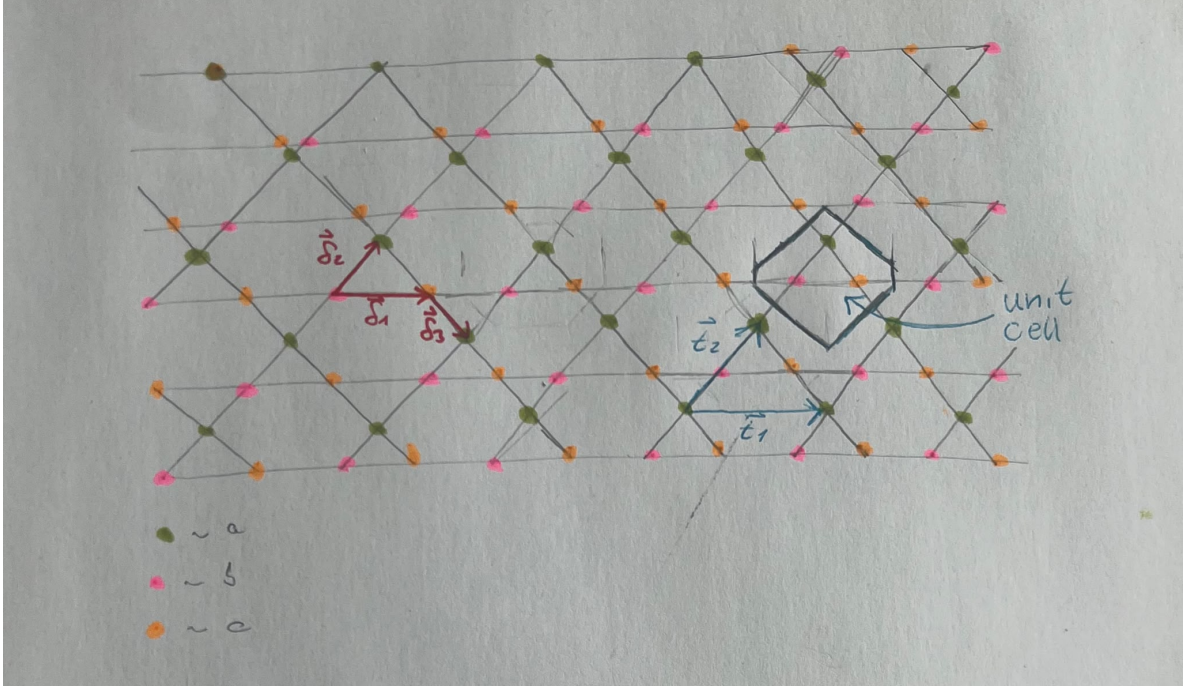


0.1 0-flux State

0.1.1 Real Space

The bare Kagome lattice is a triangular Bravais lattice with a three site unit cell (called a, b and c-sites below). The Wigner Seitz-cell is a hexagon (with full symmetries, even if the drawing suggests otherwise ;)). Geometrically speaking the lattice consists of corner-sharing equilateral triangles.



In all that follows I normalize all distances wrt the distance between nearest neighbours. The nearest neighbour vectors are (see cartoon)

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \quad (1)$$

and the primitive translations

$$\vec{t}_1 = 2\vec{\delta}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = 2\vec{\delta}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}. \quad (2)$$

The tight binding hamiltonian is

$$H = t \sum_{i\sigma} \left(a_{i\sigma}^\dagger c_{i\sigma} + c_{i\sigma}^\dagger b_{i\sigma} + b_{i\sigma}^\dagger a_{i\sigma} + c_{i\sigma}^\dagger b_{i+t_1;\sigma} + c_{i\sigma}^\dagger a_{i+t_1-t_2;\sigma} + a_{i\sigma}^\dagger b_{i+t_2;\sigma} + \text{h.c.} \right) \quad (3)$$

The first index refers to the unit cell and the second one to the "species" of the particle, the latter being a product of the parton decomposition.

0.1.2 Reciprocal Space

The reciprocal lattice vectors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix} \quad (4)$$

The first Brillouin zone is also a hexagon (rotated by 90° wrt to the Wigner-Seitz cell). To find H in reciprocal space I do a Fourier transform:

$$a_{i\sigma} = \sqrt{\frac{3}{N}} \sum_k e^{i\vec{k}\cdot\vec{R}_i} a_{k\sigma}, \quad (5)$$

where N is the number of unit cells. This yields

$$H = 2t \sum_{q\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger & c_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & \cos(\vec{q} \cdot \vec{\delta}_2) & \cos(\vec{q} \cdot \vec{\delta}_3) \\ \cos(\vec{q} \cdot \vec{\delta}_2) & 0 & \cos(\vec{q} \cdot \vec{\delta}_1) \\ \cos(\vec{q} \cdot \vec{\delta}_3) & \cos(\vec{q} \cdot \vec{\delta}_1) & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix}. \quad (6)$$

0.1.3 dispersion and band touching points

The three eigenvalues are

$$\varepsilon_1(\vec{q}) = t \left(1 - \sqrt{4 \cos^2(q_x) + 4 \cos(q_x) \cos(\sqrt{3}q_y) + 1} \right) \quad (7)$$

$$\varepsilon_2(\vec{q}) = t \left(1 + \sqrt{4 \cos^2(q_x) + 4 \cos(q_x) \cos(\sqrt{3}q_y) + 1} \right) \quad (8)$$

$$\varepsilon_3(\vec{q}) = -2t, \quad (9)$$

see [1]. I also used

$$\cos^2\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) + \cos^2\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) = \cos(q_x) \cos(\sqrt{3}q_y) + 1 \quad (10)$$

Interestingly, the top band is completely flat. The other two bands touch when

$$\cos^2(q_x) + \cos(q_x) \cos(\sqrt{3}q_y) + 1/4 = 0 \quad (11)$$

$$\Rightarrow \cos(q_x) = \frac{1}{2} \left(-\cos(\sqrt{3}q_y) \pm \sqrt{\cos^2(\sqrt{3}q_y) - 1} \right) \quad (12)$$

$$\Rightarrow q_y = \frac{\pi l}{\sqrt{3}}, \quad l \in \mathbb{N} \quad (13)$$

$$\Rightarrow q_x = (-1)^s \left(\frac{\pi}{2} + \frac{(-1)^l \pi}{6} \right) + 2\pi m, \quad m \in \mathbb{N}, \quad s \in [0, 1] \quad (14)$$

A handy solution to this equation is

$$\begin{aligned} q_x &= (-1)^s \frac{2\pi}{3} + m_1 \pi \\ q_y &= \frac{\pi}{\sqrt{3}} (2m_2 - m_1), \end{aligned} \quad (15)$$

with $s \in 0, 1$ and $m_1, m_2 \in \mathbb{N}$. These touching points lie at the corners of the Brillouin zone, they come in two flavours. The parameter s labels the aforementioned flavours while m_1 (m_2) is a translation by the reciprocal lattice vector $\vec{b}_1$ ($\vec{b}_2$). For example

$$\vec{K} = \begin{pmatrix} 2\pi/3 \\ 0 \end{pmatrix}, \quad s = 0, \quad m_1 = 0, \quad m_2 = 0 \quad (16)$$

$$\vec{K}' = \begin{pmatrix} -2\pi/3 \\ 0 \end{pmatrix}, \quad s = 1, \quad m_1 = 0, \quad m_2 = 0. \quad (17)$$

0.1.4 Filling Factor

The idea is the following: The lower two bands contain the same number of states by symmetry. By calculating the number of states of the upper band (should be easy) it should be possible to calculate at what filling factor the low energy physics happen exactly the linear touching points.

0.1.5 Linearization in Reciprocal Space and Projection

Use

$$\cos\left(\left(\vec{K}^{(\prime)} + \vec{p}\right) \cdot \vec{\delta}_{ij}\right) \approx \cos\left(\vec{K}^{(\prime)} \cdot \vec{\delta}_{ij}\right) - \sin\left(\vec{K} \cdot \vec{\delta}_{ij}\right) \left(\vec{p} \cdot \vec{\delta}_{ij}\right) \quad (18)$$

and

$$\begin{aligned} \vec{K} \cdot \vec{\delta}_1 &= \frac{2\pi}{3} \\ \vec{K} \cdot \vec{\delta}_2 &= \frac{\pi}{3} \\ \vec{K} \cdot \vec{\delta}_3 &= \frac{\pi}{3} \\ \vec{K}' &= -\vec{K} \end{aligned} \quad (19)$$

$$H_{\vec{K}} \approx t \sum_{p\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger & c_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_3 \\ 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 0 & -1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_1 \\ 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_3 & -1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_1 & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix} \quad (20)$$

Directly at the $\vec{K}$ -point the eigenvalues are obviously

$$\varepsilon_1(\vec{K}) = t, \quad \varepsilon_2(\vec{K}) = t, \quad \varepsilon_3(\vec{K}) = -2t, \quad (21)$$

the eigenvector corresponding to the flat band reads at the $\vec{K}$ -point:

$$\vec{v}_3(\vec{K}) = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}. \quad (22)$$

Since H is hermitian (/symmetric in this case) there exist a orthonormal basis of eigenvectors. Obviously the subspace orthonormal to $\vec{v}_3(\vec{K})$ corresponds to the two bands that cross. A possible (orthonormal) basis for this subspace is

$$\vec{v}_1(\vec{K}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{v}_2(\vec{K}) = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}. \quad (23)$$

Projecting H onto this basis yields

$$\begin{aligned} \vec{v}_1^T \cdot H_{\vec{K}} \cdot \vec{v}_1 &= t - \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} & \vec{v}_1^T \cdot H_{\vec{K}} \cdot \vec{v}_2 &= -t(\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p} \\ \vec{v}_2^T \cdot H_{\vec{K}} \cdot \vec{v}_1 &= -t(\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p} & \vec{v}_2^T \cdot H_{\vec{K}} \cdot \vec{v}_2 &= t + \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} \end{aligned}$$

$$\Leftrightarrow t\sigma_0 - \sqrt{3}t(\vec{\delta}_2 \cdot \vec{p})\sigma_z + t((\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p})\sigma_x \quad (24)$$

This already has the correct dispersion

$$\varepsilon_{1,2}(\vec{K} + \vec{p}) \approx \sqrt{3}t|\vec{p}|, \quad (25)$$

but is not yet in the form of the Weyl equation. To transform the equation into the Weyl equation one can rotate the basis inside the degenerate subspace. I take the following approach; define

$$\begin{aligned}\vec{w}_1 &= \alpha \vec{v}_1 + \beta \vec{v}_2 \\ \vec{w}_2 &= \eta \vec{v}_1 + \gamma \vec{v}_2\end{aligned}\tag{26}$$

Othogonality dictates

$$\begin{aligned}|\alpha|^2 + |\beta|^2 &= 1 \\ |\eta|^2 + |\gamma|^2 &= 1 \\ \alpha^* \eta + \beta^* \gamma &= 0.\end{aligned}$$

Maybe make an Ansatz using trig functions and phases? Though I should count free parameters before. Educated guess with 3 free parameters:

$$\begin{aligned}\alpha &= e^{i\theta_1} \sin(\phi) \\ \beta &= e^{i\theta_2} \cos(\phi) \\ \eta &= e^{i\theta_1} \cos(\phi) \\ \gamma &= -e^{i\theta_2} \sin(\phi).\end{aligned}$$

Utilizing $\vec{\delta}_2 + \vec{\delta}_3 = \vec{\delta}_1$ I get:

$$\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_1 = t + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(1 - 2|\alpha|^2 - \frac{2\text{Re}(\alpha^* \beta)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{4\text{Re}(\alpha^* \beta)}{\sqrt{3}} \right) \tag{27}$$

$$\vec{w}_2^H \cdot H_{\vec{K}} \cdot \vec{w}_2 = 1 + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(1 - 2|\eta|^2 - \frac{2\text{Re}(\eta^* \gamma)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{4\text{Re}(\eta^* \gamma)}{\sqrt{3}} \right) \tag{28}$$

$$\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_2 = \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(-2\alpha^* \eta - \frac{\alpha^* \gamma + \beta^* \eta}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{2(\alpha^* \gamma + \beta^* \eta)}{\sqrt{3}} \right) \tag{29}$$

$$\vec{w}_2^H \cdot H_{\vec{K}} \cdot \vec{w}_1 = (\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_2)^* \tag{30}$$

The strategy is to solve for values for α, β, γ and η such that the following combinations occur in the matrix elements:

$$p_x = (\vec{\delta}_2 + \vec{\delta}_3) \cdot \vec{p} \tag{31}$$

$$p_y = \frac{1}{\sqrt{3}} (\vec{\delta}_2 - \vec{\delta}_3) \cdot \vec{p} \tag{32}$$

whil still leading to orthonormal basis vectors. The following parameters are a sensible choice

$$\alpha = -\frac{1}{2}, \quad \beta = \frac{\sqrt{3}}{2}, \quad \eta = \frac{\sqrt{3}}{2}, \quad \gamma = \frac{1}{2}. \tag{33}$$

Thus, the new basis vectors read

$$\vec{w}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad \vec{w}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \tag{34}$$

$$\Rightarrow \langle H_{\vec{K}} \rangle_{\vec{w}_1, \vec{w}_2} = t\sigma_0 + \sqrt{3}tp_y\sigma_x + \sqrt{3}tp_x\sigma_z \tag{35}$$

This form was also found in [2], a different one was obtained in [3]. I expect that I would find the latter if I chose different values for α, β, γ and η . I decided to stick with this form to keep the expressions real.

I could eliminate the σ_0 term by introducing chemical potentials but I guess for non-interacting particles it doesn't really matter.

Question: when rotating the orthonormal basis inside the degenerate subspace there is one free parameter (the rotation angle). On the other hand one can probably also think about the surface of a Bloch sphere (the radius is obviously fixed). What is the free parameter in this picture /what is the fixed parameter? It is probably worthwhile to explore this at some point, it might reveal the underlying topology.

Addition: the counting of free coefficients might not be correct, considering they can be complex. Is the Bloch sphere argument even valid?

0.1.6 Introduction of Strain

I show how to expand expressions prototypical for the k-space Hamiltonian. First I expand in strain:

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{q}\cdot\vec{\delta}_{ij}} \left(1 - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{q} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \right), \quad (36)$$

afterwards I expand around $\vec{K}$:

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{q}\cdot\vec{\delta}_{ij}} \left(1 + \vec{p} \cdot \vec{\delta}_{ij} \right) \left(1 - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\underline{u} \cdot \vec{\delta}_{ij} \right) \cdot \vec{p} \right). \quad (37)$$

This compares well with the field theory method for the triangular lattice where I did the expansions in reverse order.

Keeping only the relevant terms (relevant to this order at least):

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{K}\cdot\vec{\delta}_{ij}} \left(1 + i\vec{p} \cdot \vec{\delta}_{ij} - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \right) \quad (38)$$

$$\begin{aligned} \Rightarrow t(-\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} + t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} &\approx 2t \cos(\vec{K} \cdot \vec{\delta}_{ij}) - 2t \sin(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{p} \cdot \vec{\delta}_{ij} \right) \\ &\quad - 2\beta t \cos(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \\ &\quad - 2t \sin(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \end{aligned} \quad (39)$$

I group the linear corrections by order in β to find the two matrices:

$$\delta H_1 = -2t\beta \begin{pmatrix} 0 & (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy})/8 \\ (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & 0 & -u_{xx}/2 \\ (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy})/8 & -u_{xx}/2 & 0 \end{pmatrix} \quad (40)$$

$$\delta H_2 = -2t \begin{pmatrix} 0 & \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 \\ \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & 0 & \pi\frac{u_{xx}}{\sqrt{3}} \\ \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 & \pi\frac{u_{xx}}{\sqrt{3}} & 0 \end{pmatrix} \quad (41)$$

The projections are

$$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2} = -t\beta \begin{pmatrix} u_{xx} & u_{yx} \\ u_{yx} & u_{yy} \end{pmatrix} \quad (42)$$

$$\langle \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = -\frac{2\pi t}{\sqrt{3}} \begin{pmatrix} u_{xx} & u_{yx} \\ u_{yx} & -u_{xx} \end{pmatrix} = -\frac{2\pi t}{\sqrt{3}} (u_{yx}\sigma_y + u_{xx}\sigma_z) \quad (43)$$

$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2}$ can be written as vector potential + scalar potential using:

$$\begin{aligned} u_{xx} &= \frac{1}{2} (u_{xx} - u_{yy}) + \frac{1}{2} (u_{xx} + u_{yy}) \\ u_{yy} &= -\frac{1}{2} (u_{xx} - u_{yy}) + \frac{1}{2} (u_{xx} + u_{yy}). \end{aligned} \quad (44)$$

$$\Rightarrow \langle H_{\vec{K}} + \delta H_1 + \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = (t + \phi)\sigma^0 + \sqrt{3}t \left(p_y + A_{\vec{K}}^y \right) \sigma_x + \sqrt{3}t \left(p_x + A_{\vec{K}}^x \right) \sigma_z \quad (45)$$

$$\phi = \frac{t}{2} (u_{xx} + u_{yy}) \quad (46)$$

$$A_{\vec{K}}^x = -\frac{t\beta}{2\sqrt{3}} (u_{xx} - u_{yy}) - \frac{2\pi t}{3} u_{xx} \quad (47)$$

$$A_{\vec{K}}^y = -\frac{t\beta}{\sqrt{3}} u_{yx} - \frac{2\pi t}{3} u_{yx} \quad (48)$$

The β -independent part of $\vec{A}_{\vec{K}}$ can be eliminated using a gauge-transformation. This agrees with the results here [3]. **It bothers me that only u_{yx} appears but not u_{xy} . Check this**

I give a few intermediate results for the derivation of $\vec{A}_{\vec{K}'}$:

The cool kids introduce a cone flavour index and derive the expressions for both flavours at the same time

$$H_{\vec{K}'} \approx t \sum_{p\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger & c_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & 1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_3 \\ 1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 0 & -1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_1 \\ 1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_3 & -1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_1 & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix} \quad (49)$$

(the zeroth order is the same so I make the same Ansatz for the initial basis.)

$$\begin{aligned} \vec{v}_1^T \cdot H_{\vec{K}'} \cdot \vec{v}_1 &= t + \sqrt{3}t \vec{\delta}_2 \cdot \vec{p} & \vec{v}_1^T \cdot H_{\vec{K}'} \cdot \vec{v}_2 &= t \left(\vec{\delta}_1 + \vec{\delta}_3 \right) \cdot \vec{p} \\ \vec{v}_2^T \cdot H_{\vec{K}'} \cdot \vec{v}_1 &= t \left(\vec{\delta}_1 + \vec{\delta}_3 \right) \cdot \vec{p} & \vec{v}_2^T \cdot H_{\vec{K}'} \cdot \vec{v}_2 &= t - \sqrt{3}t \vec{\delta}_2 \cdot \vec{p} \end{aligned}$$

$$\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_1 = t + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(-1 + 2|\alpha|^2 + \frac{2\text{Re}(\alpha^*\beta)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(\frac{4\text{Re}(\alpha^*\beta)}{\sqrt{3}} \right) \quad (50)$$

$$\vec{w}_2^H \cdot H_{\vec{K}'} \cdot \vec{w}_2 = 1 + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(-1 + 2|\eta|^2 + \frac{2\text{Re}(\eta^*\gamma)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(\frac{4\text{Re}(\eta^*\gamma)}{\sqrt{3}} \right) \quad (51)$$

$$\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_2 = \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(2\alpha^*\eta + \frac{\alpha^*\gamma + \beta^*\eta}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(\frac{2(\alpha^*\gamma + \beta^*\eta)}{\sqrt{3}} \right) \quad (52)$$

$$\vec{w}_2^H \cdot H_{\vec{K}'} \cdot \vec{w}_1 = (\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_2)^* \quad (53)$$

$$\Rightarrow \alpha = \frac{\sqrt{3}}{2}, \beta = \frac{1}{2}, \eta = \frac{1}{2}, \gamma = -\frac{\sqrt{3}}{2} \quad (54)$$

$$\Rightarrow \vec{w}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \vec{w}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad (55)$$

These are the same basis vectors as for $\vec{K}$ just in reverse order.

$$\Rightarrow \langle H_{\vec{K}'} \rangle_{\vec{w}_1, \vec{w}_2} = t\sigma_0 + \sqrt{3}tp_y\sigma_x + \sqrt{3}tp_x\sigma_z \quad (56)$$

δH_1 and δH_2 are the same as for $\vec{K}'$ (see 39 and substitute $\vec{K} \rightarrow \vec{K}' = \vec{K}$). With the reversed order of the basis vectors this gives

$$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2} = -t\beta \begin{pmatrix} u_{yy} & u_{yx} \\ u_{yx} & u_{xx} \end{pmatrix} \quad (57)$$

$$\langle \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = -\frac{2\pi t}{\sqrt{3}} \begin{pmatrix} -u_{xx} & u_{yx} \\ u_{yx} & u_{xx} \end{pmatrix} = -\frac{2\pi t}{\sqrt{3}} (u_{yx}\sigma_y - u_{xx}\sigma_z) \quad (58)$$

$$\phi = \frac{t}{2}(u_{xx} + u_{yy}) \quad (59)$$

$$A_{\vec{K}'}^x = \frac{t\beta}{2\sqrt{3}}(u_{xx} - u_{yy}) + \frac{2\pi t}{3}u_{xx} \quad (60)$$

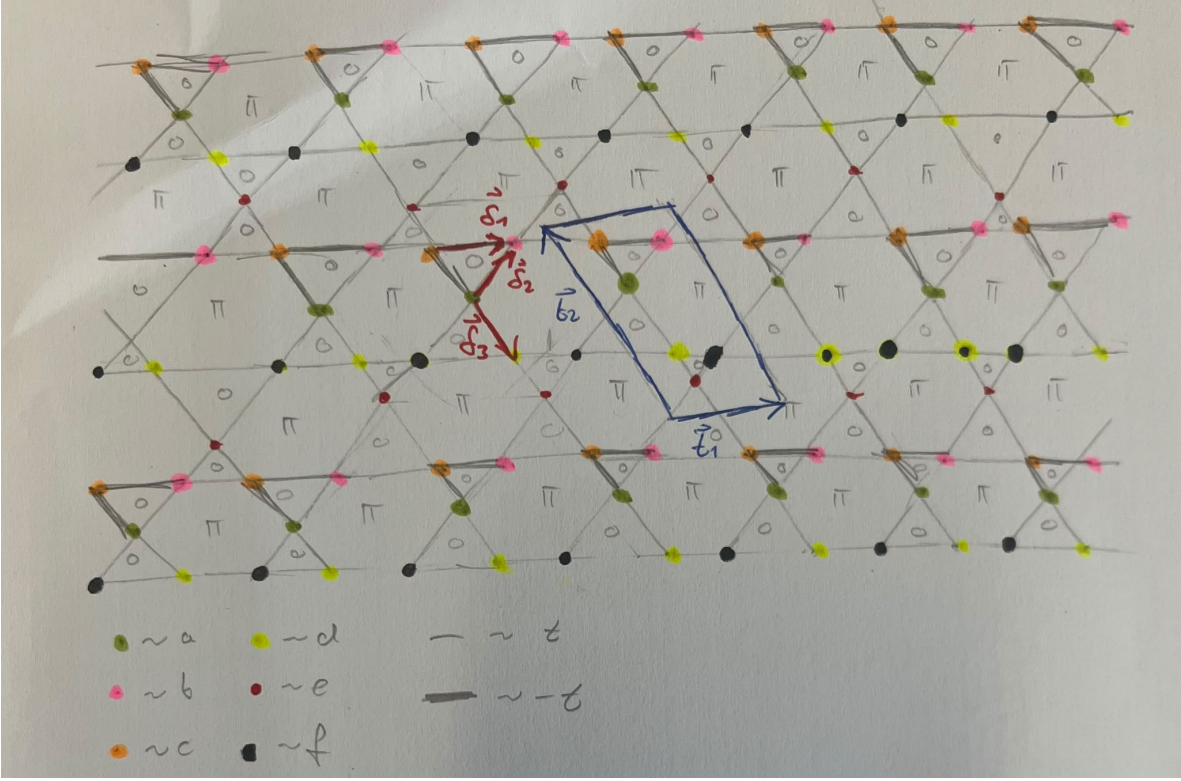
$$A_{\vec{K}'}^y = \frac{t\beta}{\sqrt{3}}u_{yx} + \frac{2\pi t}{3}u_{yx} \quad (61)$$

$$\Rightarrow A_{\vec{K}} = -A_{\vec{K}'} \quad (62)$$

0.2 π -flux State

0.2.1 Real Space

This system is a monoclinic (Is this correct?) Bravais-lattice with a six site unit cell. It emerges from the bare Kagome lattice through unit cell doubling (the translational symmetry gets broken by the internal mean field).



The nearest neighbour vectors remain unchanged

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \quad (63)$$

The new primitive translations are

$$\vec{t}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = \begin{pmatrix} -2 \\ 2\sqrt{3} \end{pmatrix} \quad (64)$$

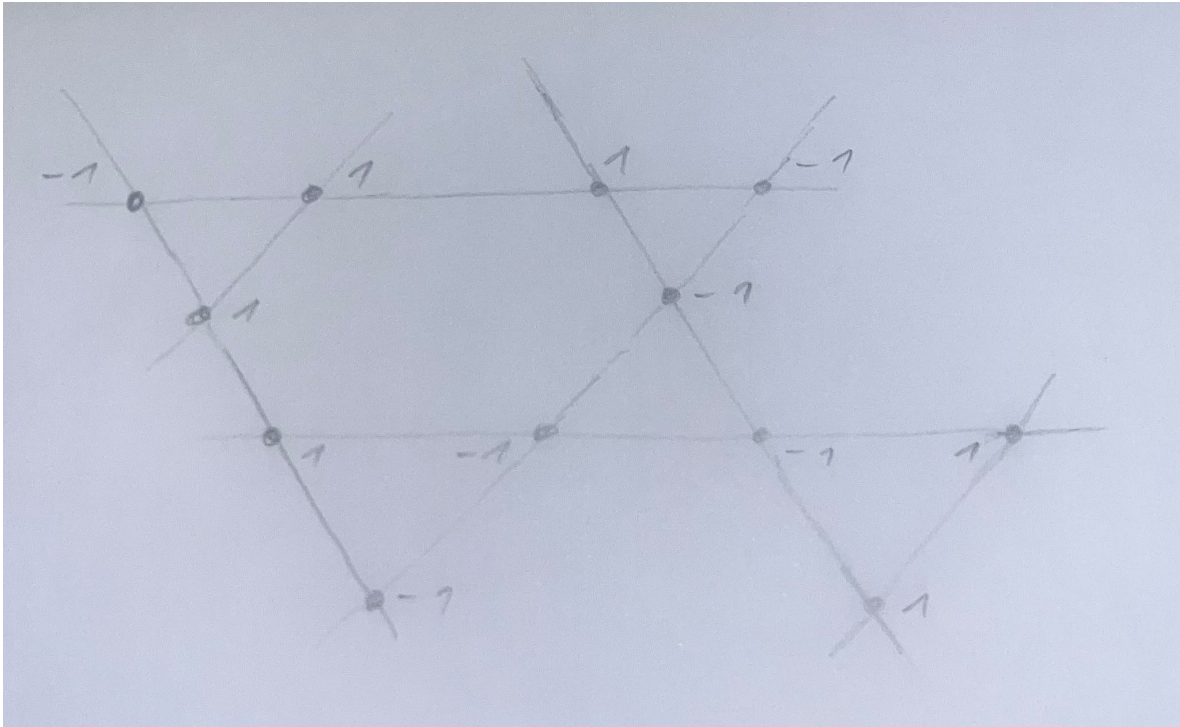
The real space (effective) tight-binding hamiltonian is

$$H = t \sum_{i\sigma} \left(-a_{i\sigma}^\dagger c_{i\sigma} - c_{i\sigma}^\dagger b_{i\sigma} + b_{i\sigma}^\dagger a_{i\sigma} + a_{i\sigma}^\dagger d_{i\sigma} + d_{i\sigma}^\dagger e_{i\sigma} + e_{i\sigma}^\dagger f_{i\sigma} + f_{i\sigma}^\dagger d_{i\sigma} \right. \\ \left. + b_{i\sigma}^\dagger c_{i+t_1;\sigma} + b_{i\sigma}^\dagger e_{i+t_1+t_2;\sigma} + f_{i\sigma}^\dagger d_{i+t_1;\sigma} + f_{i\sigma}^\dagger a_{i+t_1;\sigma} + e_{i\sigma}^\dagger c_{i-t_2;\sigma} + \text{h.c.} \right) \quad (65)$$

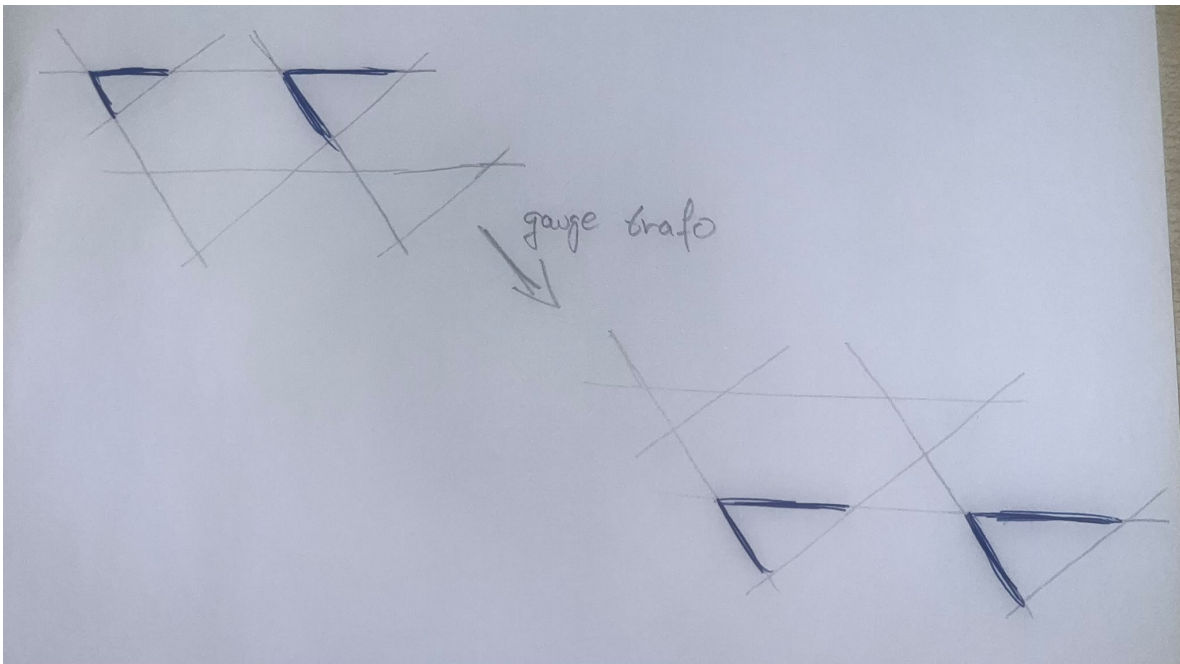
0.2.2 Symmetries

On the level of gauge-invariant quantities (that is plaquette fluxes), the system does not break any lattice symmetries, that the bare Kagome lattice has.

The specific gauge choice however looks like it lowers the translational symmetry in one direction ($\vec{t}_2$). Obviously this is a problem of the gauge-choice, the correct symmetries are dictated by the gauge-independent observables. To "repair" this one can supplement lattice symmetry transformations with gauge transformations, as gauge transformations are always allowed. This whole idea was formally introduced here: [4]. Specifically one can translate the system by $\vec{t}_1/2$ (the symmetry operation of the bare kagome lattice) followed by a gauge transformation. The correct gauge transformation in this case is



This has the effect



This then reinstates the correct symmetry. So much for the idea. No for the technical implementation. I define lattice translations

$$T_{\vec{d}} \equiv e^{i\vec{d} \cdot \vec{q}}, \quad (66)$$

,where $\vec{q}$ is the momentum operator (What is the correct momentum operator?), and a gauge transformations

$$G_{\{\theta_i\}} \equiv e^{i \sum_i \theta_i n_i}. \quad (67)$$

What follows is a plethora of questions

- What is the point group of the bare kagome lattice?
- Why are the particle numbers generators of the gauge transformations? (What is the connection to the particle number constraint?)
- Do particle number and momentum commute? How would I even translate momentum to occupation numbers?
- The description using momentum as quantum numbers must be still correct, no?
- What are the correct quantum numbers considerin the projective symmetry?
- Can one still think about the resciprocal space and which modifications are needed?
- How can one imagine the eigenfunctions of e.g. $G_{\{\theta_i\}} \cdot T_{\vec{t}/2}$?
- The band structure must be gauge-invariant, no?
- How do the bands map under the projective symmetry?

Looking further ahead at strain, can you maybe argue that this forbids pseudo magnetic fields?

0.2.3 Reciprocal Space

The reciprocal lattice vectrors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix} \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix} \quad (68)$$

After a Fourier transform the k -space Hamiltonian reads

$$H = t \sum_{q,\sigma} \Psi_{q\sigma}^\dagger \begin{pmatrix} 0 & e^{iq_2} & -e^{-iq_3} & 0 & e^{-iq_2} & e^{iq_3} \\ e^{-iq_2} & 0 & 2i \sin(q_1) & e^{iq_2} & 0 & 0 \\ -e^{iq_3} & -2i \sin(q_1) & 0 & e^{-iq_3} & 0 & 0 \\ 0 & e^{-iq_2} & e^{iq_3} & 0 & e^{iq_2} & e^{-iq_3} \\ e^{iq_2} & 0 & 0 & e^{-iq_2} & 0 & 2 \cos(q_1) \\ e^{-iq_3} & 0 & 0 & e^{iq_3} & 2 \cos(q_1) & 0 \end{pmatrix} \Psi_{q\sigma}, \quad (69)$$

where

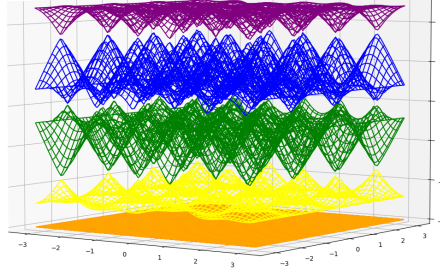
$$\Psi_{q\sigma}^\dagger = (a_{q\sigma}^\dagger \quad b_{q\sigma}^\dagger \quad c_{q\sigma}^\dagger \quad e_{q\sigma}^\dagger \quad f_{q\sigma}^\dagger \quad d_{q\sigma}^\dagger). \quad (70)$$

If I write this down somewhere serious, I should just label the sites differently.

I used the shorthand $q_i \equiv \vec{q} \cdot \vec{\delta}_i$.

0.2.4 dispersion and band trouching poiunts

The dispersion relation is shwon in the following picture



This plot was generated by numerically calculating the eigenvalues of H on a (q_x, q_y) -grid. The lowest band (flat band) is doubly degenerate.

The characteristic polynomial of 69 is (this was generated by Mathematica)

$$x^6 - 12x^4 - 8x^3 + x^2(30 + 2\Delta_{\vec{q}}) + x(24 + 8\Delta_{\vec{q}}) + 8\Delta_{\vec{q}} - 8 \quad (71)$$

units! with

$$\Delta_{\vec{q}} = -\cos(4q_2) + \cos(4q_3) - \cos(4q_1). \quad (72)$$

I should verify this using Cayley-Hamilton. To derive analytical expressions for the bands I first divide out the two flat bands (i.e. polynomial division by $(x+2)^2$). The result is

$$x^4 - 4x^3 + 8x - 2 + 2\Delta_{\vec{q}}. \quad (73)$$

To make progress I postulate (judging from the numerical plot of the dispersion) that the above polynomial can be written as a product of two quadratic polynomials. Furthermore the solutions of said quadratic equations must be centered around t . The p-q formula yields that the linear coefficients of both quadratic polynomials must therefore be $-2t$.

$$\begin{aligned} \Rightarrow x^4 - 4x^3 + 8x - 2 + 2\Delta_{\vec{q}} &= (x^2 - 2x + a) \cdot (x^2 - 2x + b) \\ &= x^4 - 4x^3 + x^2(4 + a + b) - 2x(a + b) + ab. \end{aligned} \quad (74)$$

$$\Rightarrow a + b \stackrel{!}{=} -4, \quad ab \stackrel{!}{=} -2 + 2\Delta_{\vec{q}}. \quad (75)$$

These equations are solved by

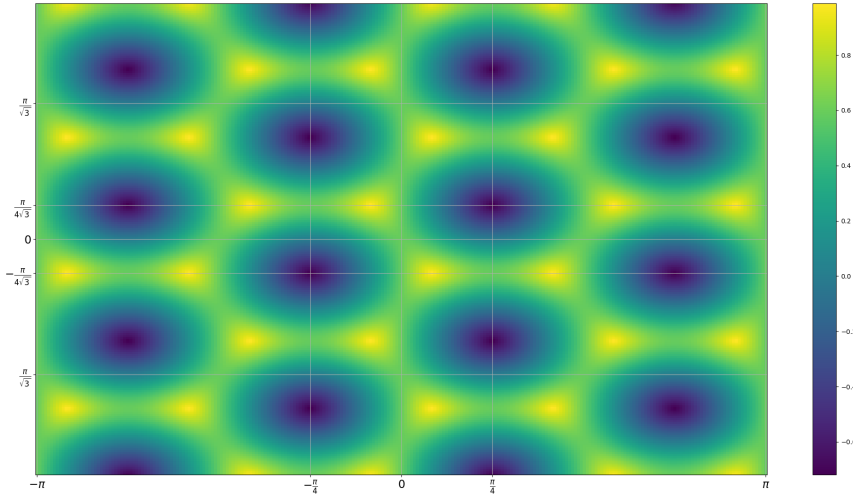
$$\begin{aligned} a_{1,2} &= -2 \pm \sqrt{6 - 2\Delta_{\vec{q}}} \\ b_{1,2} &= -2 \mp \sqrt{6 - 2\Delta_{\vec{q}}}. \end{aligned} \quad (76)$$

Both solutions lead to the same result (a and b are exchanged in the second solution wrt the first one). This gives the following expressions for the bands

$$\begin{aligned}
\varepsilon_1(\vec{q}) &= -2t \\
\varepsilon_2(\vec{q}) &= -2t \\
\varepsilon_3(\vec{q}) &= \left(1 - \sqrt{3 + \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_4(\vec{q}) &= \left(1 - \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_5(\vec{q}) &= \left(1 + \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_6(\vec{q}) &= \left(1 + \sqrt{3 + \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t.
\end{aligned} \tag{77}$$

$$\Delta_{\vec{q}} = -\cos(4q_2) + \cos(4q_3) - \cos(4q_1). \tag{78}$$

At half filling the lowest three bands are filled, thus the Fermi-energy is between the yellow and green band.



Shown above is the 4th band (green band). Apparently there are two flavours of touching points with one of each in the first Brillouin zone. They are time reversal partners. From the numerical data these two touching points lie at

$$\vec{K} = \frac{\pi}{4} \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{K}' = -\frac{\pi}{4} \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}. \tag{79}$$

Now I calculate the various touching points. It is clear from the expressions above that bands 3 and 4 as well as bands 5 and 6 touch iff

$$\begin{aligned}
6 - 2\Delta_{\vec{q}} &= 0 \\
\Leftrightarrow -\cos(4q_2) + \cos(4q_3) - \cos(4q_1) &= 3 \\
\Leftrightarrow 2\sin(2q_x)\sin(2\sqrt{3}q_y) - \cos(4q_x) &= 3.
\end{aligned} \tag{80}$$

This can easily be solved considering the ranges of sine and cosine. A possible parametrization of the solution is

$$q_x = (-1)^s \frac{\pi}{4} + m_1 \pi \quad (81)$$

$$q_y = (-1)^s \frac{\pi}{4\sqrt{3}} + (m_1 + m_2) \frac{\pi}{\sqrt{3}} \quad (82)$$

On the other hand, bands 4 and 5 touch when

$$\begin{aligned} \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}} &= 0 \\ \Leftrightarrow 2\Delta_{\vec{q}} &= -3 \\ \Leftrightarrow 4\sin(2q_x)\sin(2\sqrt{3}q_y) - 2\cos(4q_x) &= -3 \end{aligned} \quad (83)$$

Using $\cos(2x) = 1 - 2\sin^2(x)$ I obtain

$$\begin{aligned} \sin^2(2q_x) + \sin(2q_x)\sin(2\sqrt{3}q_y) + \frac{1}{4} &= 0 \\ \Rightarrow \sin(2q_x) &= \frac{1}{2} \left(-\sin(2\sqrt{3}q_y) \pm \sqrt{\sin^2(2\sqrt{3}q_y) - 1} \right) \end{aligned} \quad (84)$$

The expression under the square root gives a condition for q_y . It turns out there are two distinct sets of solutions (meaning there are two distinct sets of equivalent touching points).

$$q_x = (-1)^s \frac{\pi}{12} + m_1 \pi \quad (85)$$

$$q_y = (-1)^{s+1} \frac{\pi}{4\sqrt{3}} + (m_1 + m_2) \frac{\pi}{\sqrt{3}} \quad (86)$$

and

$$q_x = (-1)^s \frac{5\pi}{12} + m_1 \pi \quad (87)$$

$$q_y = (-1)^s \frac{\sqrt{3}\pi}{4} + (m_1 + m_2) \frac{\pi}{\sqrt{3}} \quad (88)$$

0.2.5 Linearization in reciprocal space and Projection

For now I only do the calculations for $\vec{K}$. Use

$$\vec{K} \cdot \vec{\delta}_1 = \frac{\pi}{4} \quad (89)$$

$$\vec{K} \cdot \vec{\delta}_2 = \frac{\pi}{4} \quad (90)$$

$$\vec{K} \cdot \vec{\delta}_3 = 0 \quad (91)$$

to obtain the linearized Hamiltonian in the vicinity of $\vec{K}$:

$$\begin{pmatrix} 0 & e^{i\frac{\pi}{4}}(1+ip_2) & -1+ip_3 & 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 1+ip_3 \\ e^{-i\frac{\pi}{4}}(1-ip_2) & 0 & \sqrt{2}i(1+p_1) & e^{i\frac{\pi}{4}}(1+ip_2) & 0 & 0 \\ -1-ip_3 & -\sqrt{2}i(1+p_1) & 0 & 1-ip_3 & 0 & 0 \\ 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 1+ip_3 & 0 & e^{i\frac{\pi}{4}}(1+ip_2) & 1-ip_3 \\ e^{i\frac{\pi}{4}}(1+ip_2) & 0 & 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 0 & \sqrt{2}(1-p_1) \\ 1-ip_3 & 0 & 0 & 1+ip_3 & \sqrt{2}(1-p_1) & 0 \end{pmatrix} \quad (92)$$

To keep the size reasonable maybe its a good idea to split this up into zeroth and first order in p . Might also be a good choice for storytelling purposes.

The shorthand means

$$p_i \equiv \vec{\delta}_i \cdot \vec{p},$$

also

$$e^{i\frac{\pi}{4}} = \frac{1+i}{\sqrt{2}}. \quad (93)$$

The eigenvalues for the zeroth order (/directly at the $\vec{K}$ -point) are

$$\begin{aligned} \varepsilon_1(\vec{K}) &= -2t & \varepsilon_2(\vec{K}) &= -2t & \varepsilon_3(\vec{K}) &= (1 - \sqrt{3})t \\ \varepsilon_4(\vec{K}) &= (1 - \sqrt{3})t & \varepsilon_5(\vec{K}) &= (1 + \sqrt{3})t & \varepsilon_6(\vec{K}) &= (1 + \sqrt{3})t \end{aligned} \quad (94)$$

ε_3 and ε_4 are the relevant ones.

I let mathematica calculate orthonormal eigenvectors for H at $\vec{K}$. The relevant ones (that is bands 3 and 4):

$$\vec{v}_3(\vec{K}) = \frac{1}{2\sqrt{6-3\sqrt{3}}} \left(\frac{1-\sqrt{3}}{2} + \frac{3-\sqrt{3}}{2}i, \quad \sqrt{2}(1-\sqrt{3}), \quad (-2+\sqrt{3})i, \quad \frac{1-\sqrt{3}}{2} + \frac{\sqrt{3}-3}{2}i, \quad 0, \quad 1 \right)^T \quad (95)$$

$$\vec{v}_4(\vec{K}) = \frac{1}{2\sqrt{6(7-4\sqrt{3})}} \left((3-2\sqrt{3}) + (-2+\sqrt{3})i, \quad 0, \quad i(1-\sqrt{3}), \quad 3-2\sqrt{3} + (2-\sqrt{3})i, \quad 2\sqrt{2}(2-\sqrt{3}), \quad 5-3\sqrt{3} \right)^T \quad (96)$$

The projections onto those vectors are

$$\begin{aligned} \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= (1 - \sqrt{3})t - \frac{2t}{\sqrt{3}}p_2 + \frac{(\sqrt{3}-1)t}{\sqrt{3}}p_3 \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= (1 - \sqrt{3})t + \frac{2t}{\sqrt{3}}p_2 - \frac{(\sqrt{3}-1)t}{\sqrt{3}}p_3 \\ \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= -\frac{2t}{\sqrt{3}}p_2 - \frac{(3+\sqrt{3})t}{3}p_3 \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= -\frac{2t}{\sqrt{3}}p_2 - \frac{(3+\sqrt{3})t}{3}p_3 \end{aligned} \quad (97)$$

I transform these to p_x and p_y :

$$\begin{aligned} \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= (1 - \sqrt{3})t + \frac{1-\sqrt{3}}{2}tp_x - \frac{1+\sqrt{3}}{2}tp_y \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= (1 - \sqrt{3})t - \frac{1-\sqrt{3}}{2}tp_x + \frac{1+\sqrt{3}}{2}tp_y \\ \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \end{aligned} \quad (98)$$

The dispersion of this projection is

$$\begin{aligned}\varepsilon_{3\vec{K}} &= (1 - \sqrt{3})t - \sqrt{2}t\sqrt{p_x^2 + p_y^2} \\ \varepsilon_{4\vec{K}} &= (1 - \sqrt{3})t + \sqrt{2}t\sqrt{p_x^2 + p_y^2}\end{aligned}\tag{99}$$

Most importantly

$$v_F = \sqrt{2}t\tag{100}$$

Judging from the pictures of the dispersion I didnt expect it to be so isotropical. Check this.

I follow the usual method defining

$$\begin{aligned}\vec{w}_3 &= \alpha\vec{v}_3 + \beta\vec{v}_4 \\ \vec{w}_4 &= \eta\vec{v}_3 + \gamma\vec{v}_4\end{aligned}\tag{101}$$

Orthonormality is garuanted by

$$|\alpha|^2 + |\beta|^2 = 1\tag{102}$$

$$|\eta|^2 + |\gamma|^2 = 1\tag{103}$$

$$\alpha^*\eta + \beta^*\gamma = 0\tag{104}$$

If my counting is correct, there are 8 parameters to begin with (real and imaginary parts), the equations above eliminate 1 and 1 and 2 (the last equation are really 2 equations, one for real and one for imaginary part), leaving behind 4 free parameters. I make the following Ansatz for the leftover parameters:

$$\begin{aligned}\alpha &= \cos(\phi/2)e^{i(\chi+\theta/2)}e^{i\kappa} \\ \beta &= \sin(\phi/2)e^{i(\chi+\theta/2)} \\ \eta &= \sin(\phi/2)e^{i(\chi-\theta/2)}e^{i\kappa} \\ \gamma &= -\cos(\phi/2)e^{i(\chi-\theta/2)}\end{aligned}\tag{105}$$

If you want to be fancy about it you can think of what kind of matrix describes the relation between the two bases and then find a suitable parametrization for it. If one thinks of these matrices in terms of groups, maybe there is a suffisticated argument to be made why $U(1)$ appears.. ϕ is a rotation angle, χ a global phase, θ the phase difference between the two states and κ state-internal phase. It is clear that all physical quantities can not depend on χ . At this point I might as well choose $\chi = 0$.

The following relations are impotant to derive the matrix elements in the rotated basis:

$$\begin{aligned}(2|\alpha|^2 - 1) &= \cos(\phi) \\ 2\text{Re}(\alpha^*\beta) &= \sin(\phi)\cos(\kappa) \\ \alpha^*\eta - \beta^*\gamma &= \sin(\phi)\cos(\theta) - i\sin(\phi)\sin(\theta) \\ \alpha^*\gamma + \beta^*\eta &= \sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta) \\ &\quad + i(\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta))\end{aligned}$$

Using these, in general (while neglecting the σ_0 -term),

$$\vec{w}_1^H \cdot H \cdot \vec{w}_1 \sim (\vec{v}_1^H \cdot H \cdot \vec{v}_1) \cos(\phi) + (\vec{v}_1^H \cdot H \cdot \vec{v}_2) \sin(\phi) \cos(\kappa) \quad (106)$$

$$\begin{aligned} \vec{w}_1^H \cdot H \cdot \vec{w}_2 &= (\vec{v}_1^H \cdot H \cdot \vec{v}_1) \sin(\phi) \cos(\theta) + (\vec{v}_1^H \cdot H \cdot \vec{v}_2) (\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)) \\ &\quad - i [(\vec{v}_1^H \cdot H \cdot \vec{v}_1) \sin(\phi) \sin(\theta) - (\vec{v}_1^H \cdot H \cdot \vec{v}_2) (\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta))] \end{aligned} \quad (107)$$

I find for the projection in the transformed basis (I again drop the σ_0 part as it transforms trivially):

$$\begin{aligned} \frac{\vec{w}_3^H \cdot H_{\vec{K}} \cdot \vec{w}_3}{\sqrt{2}t} &\sim \frac{1}{\sqrt{8}} \left((1 - \sqrt{3}) \cos(\phi) - (1 + \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_x \\ &\quad + \frac{1}{\sqrt{8}} \left(-(1 + \sqrt{3}) \cos(\phi) - (1 - \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_y \\ \frac{\vec{w}_4^H \cdot H_{\vec{K}} \cdot \vec{w}_4}{\sqrt{2}t} &\sim \frac{1}{\sqrt{8}} \left(-(1 - \sqrt{3}) \cos(\phi) + (1 + \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_x \\ &\quad + \frac{1}{\sqrt{8}} \left((1 + \sqrt{3}) \cos(\phi) + (1 - \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_y \\ \frac{\vec{w}_3^H \cdot H_{\vec{K}} \cdot \vec{w}_4}{\sqrt{2}t} &= \frac{1}{\sqrt{8}} \left((1 - \sqrt{3}) \sin(\phi) \cos(\theta) - (1 + \sqrt{3}) [\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)] \right) p_x \\ &\quad + \frac{1}{\sqrt{8}} \left(-(1 + \sqrt{3}) \sin(\phi) \cos(\theta) - (1 - \sqrt{3}) [\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)] \right) p_y \\ &\quad - \frac{i}{\sqrt{8}} \left((1 - \sqrt{3}) \sin(\phi) \sin(\theta) + (1 + \sqrt{3}) [\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)] \right) p_x \\ &\quad - \frac{i}{\sqrt{8}} \left(-(1 + \sqrt{3}) \sin(\phi) \sin(\theta) + (1 - \sqrt{3}) [\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)] \right) p_y \\ \frac{\vec{w}_4^H \cdot H_{\vec{K}} \cdot \vec{w}_3}{\sqrt{2}t} &= \frac{1}{\sqrt{8}} \left((1 - \sqrt{3}) \sin(\phi) \cos(\theta) - (1 + \sqrt{3}) [\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)] \right) p_x \\ &\quad + \frac{1}{\sqrt{8}} \left(-(1 + \sqrt{3}) \sin(\phi) \cos(\theta) - (1 - \sqrt{3}) [\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)] \right) p_y \\ &\quad + \frac{i}{\sqrt{8}} \left((1 - \sqrt{3}) \sin(\phi) \sin(\theta) + (1 + \sqrt{3}) [\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)] \right) p_x \\ &\quad + \frac{i}{\sqrt{8}} \left(-(1 + \sqrt{3}) \sin(\phi) \sin(\theta) + (1 - \sqrt{3}) [\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)] \right) p_y \end{aligned}$$

From these equations the $\sigma_{0,x,y,z}$ -components can be read off. I decided to solve for an equation of the form $H = \varepsilon_0 \sigma_0 + v_F (p_x \sigma_x + p_y \sigma_z)$. The corresponding angles are

$$\begin{aligned} \cos(\phi) &= -\frac{1 + \sqrt{3}}{\sqrt{8}} \quad \Rightarrow \quad \phi = \frac{11}{12} \pi \\ \cos(\kappa) &= -\frac{1 - \sqrt{3}}{\sqrt{4 - 2\sqrt{3}}} = 1 \quad \Rightarrow \quad \kappa = 0 \\ \cos(\theta) &= \frac{1 - \sqrt{3}}{\sqrt{4 - 2\sqrt{3}}} = -1 \quad \Rightarrow \quad \theta = \pi. \end{aligned} \quad (108)$$

$\sqrt{4 - 2\sqrt{3}} = \sqrt{3} - 1$:D. First I had lhs= $1 - \sqrt{3}$, and it took me around 3 hours to find the mistake.

Another solution corresponding to the form $H = \varepsilon_0 \sigma_0 + v_F(p_x \sigma_x + p_y \sigma_y)$ is

$$\begin{aligned}\phi &= -\frac{\pi}{2} \\ \kappa &= -\frac{\pi}{2} \\ \theta &= \frac{5}{12}\pi\end{aligned}\tag{109}$$

I will continue to work with this solution from now on. Typically I will employ the first solution as a sanity check.

0.2.6 Introduction of strain

The relevant equations are

$$t(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{K}\cdot\vec{\delta}_{ij}} \left(1 + i\vec{p}\cdot\vec{\delta}_{ij} - \beta \left(\vec{\delta}_{ij}\cdot\vec{u}\cdot\vec{\delta}_{ij}\right) + i \left(\vec{K}\cdot\vec{u}\cdot\vec{\delta}_{ij}\right)\right)\tag{110}$$

and

$$\begin{aligned}t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} + t(-\vec{\delta}_{ij})e^{-i\vec{K}\cdot\vec{\delta}_{ij}} &\approx 2t \cos(\vec{q}\cdot\vec{\delta}_{ij}) - 2t \sin(\vec{q}\cdot\vec{\delta}_{ij}) (\vec{p}\cdot\vec{\delta}_{ij}) \\ &\quad - 2\beta t \cos(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{\delta}_{ij}\cdot\vec{u}\cdot\vec{\delta}_{ij}) \\ &\quad - 2t \sin(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{K}\cdot\vec{u}\cdot\vec{\delta}_{ij})\end{aligned}\tag{111}$$

and

$$\begin{aligned}t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} - t(-\vec{\delta}_{ij})e^{-i\vec{q}\cdot\vec{\delta}_{ij}} &\approx 2it \sin(\vec{K}\cdot\vec{\delta}_{ij}) + 2it \cos(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{p}\cdot\vec{\delta}_{ij}) \\ &\quad - 2i\beta t \sin(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{\delta}_{ij}\cdot\vec{u}\cdot\vec{\delta}_{ij}) \\ &\quad + 2it \cos(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{K}\cdot\vec{u}\cdot\vec{\delta}_{ij})\end{aligned}\tag{112}$$

I split the everything into two parts:

$$\begin{aligned}\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2} &= -\frac{\beta}{2}(1 - \sqrt{3})(u_{xx} + u_{yy})\sigma_0 \\ \langle \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} &= \sqrt{2}\frac{\pi}{4} \left(u_{xx} + \frac{u_{yx}}{\sqrt{3}}\right) \sigma_x + \sqrt{2}\frac{\pi}{4} \left(u_{xy} + \frac{u_{yy}}{\sqrt{3}}\right) \sigma_y\end{aligned}\tag{113}$$

The second term is a gradient field and can be gauged away. This means that to this order there is no effective vector potential but only an effective scalar potential.

Doing the same calculation for K' basically means transforming the Hamiltonian

$$H \mapsto \tilde{H} \equiv THT,\tag{114}$$

with T the time reversel. Thus, the eigenvectors of $\tilde{H}$ are $T\vec{v}_1$ and $T\vec{v}_2$ (since T is its own inveres). In this case time reversel (which can be interpreted as either $\vec{K} \mapsto -\vec{K}$ or $\vec{q} \mapsto -\vec{q}$) is equivalent to complex conjugation. Due to the antilinearity of time reversel, the matrix elements are conjugated. Thus the parameters have to be adjusted.

0.2.7 crossing 5-6

Since bands 5 and 6 cross at the same point as bands 3 and 4, the expansions of the Hamiltonian are still valid. The eigenvectors corresponding to bands 5 and 6 at the K -point are

$$\vec{v}_5(\vec{K}) = \frac{1}{\sqrt{\sqrt{6} + 3\sqrt{2}}} \begin{pmatrix} -\frac{1}{\sqrt{2}} - i\frac{2+\sqrt{3}}{\sqrt{2}} & -2 - \sqrt{3} & i\sqrt{2}(1 + \sqrt{3}) & -\frac{1}{\sqrt{2}} + i\frac{2+\sqrt{3}}{\sqrt{2}} & 1 & 0 \end{pmatrix} \quad (115)$$

$$\vec{v}_6(\vec{K}) = \frac{1}{4\sqrt{3}} \begin{pmatrix} 1 + \sqrt{3} + i(1 - \sqrt{3}) & \sqrt{2} + \sqrt{6} & 0 & 1 + \sqrt{3} - i(1 - \sqrt{3}) & \sqrt{2} + \sqrt{6} & 4 \end{pmatrix} \quad (116)$$

The projections onto $\vec{v}_5(\vec{K})$ and $\vec{v}_6(\vec{K})$ are

$$\begin{aligned} \vec{v}_5^H \cdot H_{\vec{K}} \cdot \vec{v}_5 &= (1 + \sqrt{3})t + \frac{1 + \sqrt{3}}{2}tp_x - \frac{1 - \sqrt{3}}{2}tp_y \\ \vec{v}_6^H \cdot H_{\vec{K}} \cdot \vec{v}_6 &= (1 + \sqrt{3})t - \frac{1 + \sqrt{3}}{2}tp_x + \frac{1 - \sqrt{3}}{2}tp_y \\ \vec{v}_5^H \cdot H_{\vec{K}} \cdot \vec{v}_6 &= \frac{1 - \sqrt{3}}{2}tp_x + \frac{1 + \sqrt{3}}{2}tp_y \\ \vec{v}_6^H \cdot H_{\vec{K}} \cdot \vec{v}_5 &= \frac{1 - \sqrt{3}}{2}tp_x + \frac{1 + \sqrt{3}}{2}tp_y \end{aligned} \quad (117)$$

$$\Rightarrow \langle H_{\vec{K}} \rangle_{\vec{v}_5, \vec{v}_6} = t(1 + \sqrt{3})\sigma_0 + \frac{t}{2} \left((1 + \sqrt{3})p_x - (1 - \sqrt{3})p_y \right) \sigma_z + \frac{t}{2} \left((1 - \sqrt{3})p_x + (1 + \sqrt{3})p_y \right) \sigma_x \quad (118)$$

Thus, the dispersions are

$$\varepsilon_{5\vec{K}}(\vec{p}) = (1 + \sqrt{3})t - \sqrt{2}t\sqrt{p_x^2 + p_y^2} \quad (119)$$

$$\varepsilon_{6\vec{K}}(\vec{p}) = (1 + \sqrt{3})t + \sqrt{2}t\sqrt{p_x^2 + p_y^2} \quad (120)$$

Again the Fermi velocity is $v_F = \sqrt{2}t$. I use the usual method and refine it a bit. Consider the coefficients

$$H_{11} \cos(\phi) + H_{12} \sin(\phi) \cos(\kappa) \quad (\sigma_z) \quad (121)$$

$$H_{11} \sin(\phi) \cos(\theta) + H_{12} (\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)) \quad (\sigma_x) \quad (122)$$

$$H_{11} \sin(\phi) \sin(\theta) - H_{12} (\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)) \quad (\sigma_y) \quad (123)$$

I used the shorthand $H_{ii} \equiv \vec{v}_i^T \cdot H_{\vec{K}} \cdot \vec{v}_i$. Going forward I split these coefficients into p_x and p_y parts, which I should call H_{ii}^x and H_{ii}^y , such that $H_{ii} = H_{ii}^x p_x + H_{ii}^y p_y$.

Solving for an equation of the form $H = \varepsilon_0 \sigma_0 + p_x \sigma_x + p_y \sigma_y$, eq 121 directly gives

$$\cos(\kappa) = 0$$

$$\cos(\phi) = 0,$$

assuming that $H_{11}^{x/y} \neq 0$ and $H_{12}^{x/y} \neq 0$. This is not enough to fix κ and ϕ yet as both $\pi/2$ and $-\pi/2$ are valid solutions. (I lack a rigorous argument here). Equations 122 and 123 reveal

$$\tan(\theta) = -\frac{H_{12}^x}{H_{12}^y}. \quad (124)$$

In this case that means

$$\begin{aligned}\theta &= \arctan\left(-\frac{1-\sqrt{3}}{1+\sqrt{3}}\right) = \frac{\pi}{12} \\ \phi &= \pi/2 \\ \kappa &= -\pi/2\end{aligned}\tag{125}$$

This resulting strain is

$$\langle \delta H_1 \rangle_{\vec{w}_5, \vec{w}_6} = -\frac{\beta}{2}(1+\sqrt{3})(u_{xx} + u_{yy})\sigma_0\tag{126}$$

$$\langle \delta H_2 \rangle_{\vec{w}_5, \vec{w}_6} = \sqrt{2}\frac{\pi}{4}\left(u_{xx} + \frac{u_{yx}}{\sqrt{3}}\right)\sigma_x + \sqrt{2}\frac{\pi}{4}\left(u_{xy} + \frac{u_{yy}}{\sqrt{3}}\right)\sigma_y\tag{127}$$

Apparently the scalar potential scales with the energy?

0.2.8 crossing 4-5

I begin with the degeneracy point at

$$\vec{K}'' = \frac{\pi}{12} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix},\tag{128}$$

It should be easy to verify the following calculations by repeating them for $-\vec{K}''$.

I pick the orthonormal basis vectors

$$\vec{v}_4(\vec{K}'') = \frac{1}{\sqrt{6}\sqrt{5\sqrt{3}+9}} \begin{pmatrix} \frac{1+\sqrt{3}}{\sqrt{2}} - i\sqrt{2}(2+\sqrt{3}) & 2+\sqrt{3} & i\sqrt{2}(2+\sqrt{3}) & \frac{1+\sqrt{3}}{\sqrt{2}} + i\sqrt{2}(2+\sqrt{3}) & 1 & 0 \end{pmatrix}\tag{129}$$

$$\vec{v}_5(\vec{K}'') = \frac{\sqrt{2}}{\sqrt{3\sqrt{3}(1+\sqrt{3})}} \begin{pmatrix} -\frac{1}{2} & -\frac{3}{2\sqrt{2}}(1+\sqrt{3}) & i\frac{1+\sqrt{3}}{2} & -\frac{1}{2} & \frac{1+\sqrt{3}}{2\sqrt{2}} & 1 \end{pmatrix}\tag{130}$$

The projections are

$$\vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 = -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y\tag{131}$$

$$\vec{v}_5^H \cdot H_{\vec{K}} \cdot \vec{v}_5 = \frac{1+\sqrt{3}}{2}tp_x + \frac{1-\sqrt{3}}{2}tp_y\tag{132}$$

$$\vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_5 = -\frac{1-\sqrt{3}}{2}tp_x + \frac{1+\sqrt{3}}{2}tp_y\tag{133}$$

$$\Rightarrow v_F = \sqrt{2}t\tag{134}$$

This yields the angles

$$\begin{aligned}\theta &= \arctan\left(\frac{1-\sqrt{3}}{1+\sqrt{3}}\right) = -\pi/12 \\ \phi &= -\pi/2 \\ \kappa &= -\pi/2\end{aligned}\tag{135}$$

$$\Rightarrow \langle H_{\vec{K}} \rangle_{\vec{w}_1, \vec{w}_2} = \sqrt{2} t p_x \sigma_x + \sqrt{2} t p_y \sigma_y \quad (136)$$

The resulting lowest order terms are

$$\langle \delta H_1 \rangle_{\vec{w}_4, \vec{w}_5} = -\frac{\beta}{2} (u_{xx} + u_{yy}) \sigma_0 + \sqrt{2} \frac{\beta}{2\sqrt{3}} (u_{yy} - u_{xx}) \sigma_x + \sqrt{2} \frac{\beta}{2\sqrt{3}} (u_{xy} + u_{yx}) \sigma_y \quad (137)$$

$$\langle \delta H_2 \rangle_{\vec{w}_4, \vec{w}_5} = \sqrt{2} \frac{\pi}{12} (u_{xx} - \sqrt{3} u_{yx}) \sigma_x + \sqrt{2} \frac{\pi}{12} (u_{xy} - \sqrt{3} u_{yy}) \sigma_y \quad (138)$$

Since the second term can be written as a gradient field only the first term contributes to a pseudo-magnetic field. I find

$$\phi(\vec{K}'') = -\frac{\beta}{2} (u_{xx} + u_{yy}) \quad (139)$$

$$\vec{A}(\vec{K}'') = \frac{\beta}{2\sqrt{3}} \begin{pmatrix} u_{yy} - u_{xx} \\ u_{xy} + u_{yx} \end{pmatrix}. \quad (140)$$

$$\Rightarrow B(\vec{K}'') = \frac{\beta}{2\sqrt{3}} (2\partial_x \partial_y U_x + [\partial_x^2 - \partial_y^2] U_y) \quad (141)$$

For the purpose of verification I do the same calculation for an effective theory of the form $H = \varepsilon_0 \sigma_0 + p_x \sigma_x + p_y \sigma_z$. [123](#) directly gives

$$\sin(\theta) = 0 \quad (142)$$

$$\sin(\kappa) = 0. \quad (143)$$

Furthermore, from eqs [122](#) and [123](#):

$$\tan(\phi) = \sqrt{-\frac{H_{11}^x H_{12}^y}{H_{11}^y H_{12}^x}}. \quad (144)$$

This is not yet enough to fix the sign of ϕ . I'll have to return to this to fix θ and κ , the problem is in structure identical to the other effective theory above.

The right choice is thus

$$\phi = \arctan\left(\sqrt{7 + 4\sqrt{3}}\right) = \frac{5\pi}{12} \quad (145)$$

$$\kappa = 0 \quad (146)$$

$$\theta = \pi \quad (147)$$

The projections onto those vectors are

$$\langle \delta H_1 \rangle_{\vec{w}_4, \vec{w}_5} = -\frac{\beta}{2} (u_{yy} + u_{xx}) \sigma_0 + \sqrt{2} \frac{\beta}{2\sqrt{3}} (u_{yy} - u_{xx}) \sigma_x + \sqrt{2} \frac{\beta}{2\sqrt{3}} (u_{xy} + u_{yx}) \sigma_z \quad (148)$$

$$\langle \delta H_2 \rangle_{\vec{w}_4, \vec{w}_5} = ? \quad (149)$$

Now I repeat the same for the other set of degeneracy points. Among those I pick

$$\vec{K}''' = \frac{\pi}{12} \begin{pmatrix} 5 \\ 3\sqrt{3} \end{pmatrix} \quad (150)$$

I choose the vectors

$$v_4(\vec{K}''') = \frac{1}{\sqrt{6}\sqrt{9-5\sqrt{3}}} \begin{pmatrix} \frac{1-\sqrt{3}}{\sqrt{2}} - i\sqrt{2}(2-\sqrt{3}) & 2-\sqrt{3} & i\sqrt{2}(\sqrt{3}-2) & \frac{1-\sqrt{3}}{\sqrt{2}} + i\sqrt{2}(2-\sqrt{3}) & 1 & 0 \end{pmatrix} \quad (151)$$

$$v_5(\vec{K}''') = \frac{\sqrt{2}}{\sqrt{3}\sqrt{3-\sqrt{3}}} \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2\sqrt{2}}(1-\sqrt{3}) & i\frac{1-\sqrt{3}}{2} & \frac{1}{2} & -\frac{1}{2\sqrt{2}}(1-\sqrt{3}) & 1 \end{pmatrix} \quad (152)$$

The projection coefficients are

$$\begin{aligned} H_{11}^x &= -\frac{1-\sqrt{3}}{2}t \\ H_{11}^y &= \frac{1+\sqrt{3}}{2}t \\ H_{12}^x &= -\frac{1+\sqrt{3}}{2}t \\ H_{12}^y &= -\frac{1-\sqrt{3}}{2}t. \end{aligned} \quad (153)$$

For a theory of the Form $H = \varepsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_y$ the correct parameters are

$$\begin{aligned} \theta &= \arctan(2+\sqrt{3}) = \frac{5\pi}{12} \\ \kappa &= -\pi/2 \\ \phi &= \pi/2 \end{aligned} \quad (154)$$

The projections are

$$\langle \delta H_1 \rangle_{\vec{w}_4, \vec{w}_5} = -\frac{\beta}{2}(u_{xx} + u_{yy})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xx} - u_{yy})\sigma_x - \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy} + u_{yx})\sigma_y \quad (155)$$

$$\langle \delta H_2 \rangle_{\vec{w}_4, \vec{w}_5} = \sqrt{2}\frac{\pi}{12} \left(5u_{xx} + 3\sqrt{3}u_{yx} \right) \sigma_x + \sqrt{2}\frac{\pi}{12} (5u_{xy} + 3\sqrt{3}u_{yy})\sigma_y \quad (156)$$

$$\phi(\vec{K}''') = -\frac{\beta}{2}(u_{xx} + u_{yy}) \quad (157)$$

$$\vec{A}(\vec{K}''') = \frac{\beta}{2\sqrt{3}} \begin{pmatrix} u_{xx} - u_{yy} \\ -(u_{xy} + u_{yx}) \end{pmatrix} \quad (158)$$

It turns out $\vec{A}(\vec{K}'') = -\vec{A}(\vec{K}''')$. The resulting magnetic field is

$$B(\vec{K}'') = -\frac{\beta}{2\sqrt{3}} (2\partial_x\partial_y U_x + [\partial_x^2 - \partial_y^2] U_y). \quad (159)$$

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